

Topic Modeling : A comprehensive guide

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Objective

Our objective with topic modeling will be successfully mining the large pile of textual corpus where a topic model takes a collection of documents as an input and then attempts, without supervision, to uncover the underlying topics in this collection . Each topic describes a human-interpretable semantic concept. TM then provides a latent interpretable representation of documents according to the conceived topics .

What is topic modeling:

Topic modeling in natural language processing (NLP) is the text mining technique by applying the bag of words-like technique eternally that applies in the large set of texts to produce a summary set of terms that represent the collection's overall primary set of topics .

Topic models specifically identify common keywords or phrases in a text dataset and group those words under a number of topics. They aim to uncover the latent topics or themes characterizing a set of documents. In this way, topic models are a machine learning-based form of text analysis used to thematically annotate large text corpora.

We can apply this in :

- To find out the hidden theme in a collection of documents and thus provides an automatic means to organize, understand and summarize large collections of textual information and even in sentiment analysis.

- Tagging support tickets , see when you go to the local market and search for the particular item with particular details and ask the shopkeeper for that so he can tell you ,so similarly we can use topic modeling for this purpose .Topic modeling can be used to help customer service staff analyze support queries to identify the primary issues and determine any that repeatedly occur. Based on that data, they may be able to create more informative self-serve content or help customers directly.

Why it is necessary to learn

In this world of data where a average person make 1.7MB per second data on the internet , it will be very effective to learn techniques like Topic modeling to understand , summarize , analyse the sentiment of the textual data , working with text for ecommerce giants like amazon ,flipcart it is necessary to get the reviews topics enhancing their user experience , thus making profitable business .

In short, topic modeling aids businesses in:

- Performing real-time analysis on unstructured textual data
- Learn from unstructured data at scale
- Build a consistent understanding of data, regardless of its format.

If helps one to **Scalable Summarization** It allows data scientists to make sense of huge datasets (like millions of customer reviews or scientific papers) without needing to read every single word .**No Manual Labeling** As an unsupervised method, it does not require humans to pre-tag or categorize the text; the algorithm discovers the patterns on its own . **Actionable** Insights Companies use it to automatically track emerging complaints, cluster news articles, and power recommendation engines.

Types of TM

Given below is the TM hierarchy

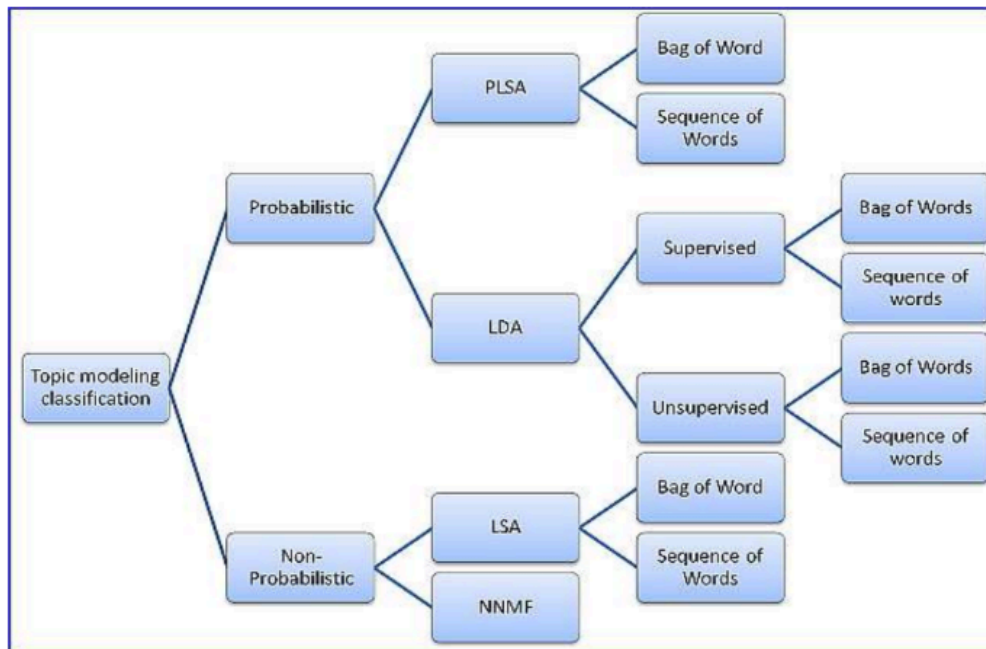


Figure 1. Topic Modeling Classification Hierarchy

our topic modeling classification strategy in this survey described as

- Probabilistic Model
- Non-Probabilistic topic model (AlgebraicModel).

In this pdf we will discuss about only two most widely used technique LSA and LDA .

Latent Semantic Analysis (LSA)

Latent Semantic Analysis (LSA) is a **natural language processing** technique used to analyze relationships between documents and the terms they contain. The method was first introduced in a paper from 1988 titled “**Using Latent Semantic Analysis to Improve Access to Textual Information**” and is still used today to create structured data from a collection of unstructured text.

Namely, LSA assumes words with similar meanings will appear in similar documents. It does so by **constructing a matrix containing** the word counts per document, where each row represents a unique word, and columns represent each document, and then using a Singular Value Decomposition (SVD) to reduce the number of rows while preserving the similarity structure among columns. SVD is a mathematical method that simplifies data while keeping its important features. It's used here to maintain the relationships between words and documents.

To determine the similarity between documents, cosine similarity is used. This is a measure that calculates the cosine of the angle between two vectors, in this case, representing documents. A

value close to 1 means the documents are very similar based on the words in them, whereas a value close to 0 means they're quite different.

Latent Dirichlet Allocation (LDA)

Latent Dirichlet Allocation (LDA) was initially proposed in 2000 in a paper titled “*Inference of population structure using multilocus genotype data*.” The paper predominantly focused on population genetics, which is a subfield of genetics concerned with genetic differences within and among populations. Three years later, **Latent Dirichlet Allocation** was applied in machine learning.

The authors of the paper describe the technique as “*a generative model for text and other collections of discrete data*.” Thus, LDA may be described as a natural language technique used to identify topics a document belongs to based on the words contained within it.

More specifically, LDA is a Bayesian network, meaning it's a generative statistical model that assumes documents are made up of words that aid in determining the topics. Thus, documents are mapped to a list of topics by assigning each word in the document to different topics. This model ignores the order of words occurring in a document and treats them as a bag of words.

LSA vs LDA

Latent Semantic Analysis (LSA) and Latent Dirichlet Allocation (LDA) are both natural language processing techniques used to create structured data from a collection of unstructured text.

However, LSA leverages Singular Value Decomposition (SVD) to reduce the dimensionality of the term-document matrix and is based on the assumption that words with similar meanings will appear in similar documents. By creating a lower-dimensional representation of the text, the model can capture the underlying relationships between words to determine how similar two documents are.

In contrast, LDA is a generative probabilistic model that leverages Bayesian inference to find the underlying topics in a corpus of texts. It assumes each document is a combination of a small number of latent topics, and each word is generated by a particular topic.

Ultimately, LSA attempts to discover the underlying relationships between words, whereas LDA seeks to discover the underlying topics in a corpus of text. Although they both are techniques used to create a vector representation of text, they make different underlying assumptions.

How it works

Topic modeling works by looking at how words cluster together across different documents. It assumes that if certain words frequently appear in the same texts, those words represent a shared theme.

Here is the exact step-by-step process of how the most common algorithm, **Latent Dirichlet Allocation (LDA)**, works behind the scenes:

1. The Core Assumptions

- **Documents are blends:** The algorithm assumes every document contains a mix of multiple topics (e.g., 60% politics, 30% economics, 10% sports).
- **Topics are blends:** It assumes every topic is a mix of specific words (e.g., the "politics" topic heavily features "election," "vote," and "government").

2. The Setup Stage

- **Clean the Text:** You feed the computer documents after removing filler words (like "the", "is", "at") and reducing words to their root form (like turning "running" into "run").
- **Set the Number:** You must tell the algorithm how many total topics (K) you want it to find in the entire collection.

3. The Initialization Stage

- **Random Guessing:** The algorithm assigns a random topic to every single word in every single document.
- **Initial Mess:** At this exact moment, the topics make absolutely no sense because the assignments are completely random.

4. The Learning Loop (Gibbs Sampling)

The algorithm runs through a continuous loop to fix its random guesses. For every word in every document, it calculates two things:

- **Topic Presence:** How common is this topic in this specific document right now?
- **Word Presence:** How common is this word across all documents assigned to this topic?

Based on these two numbers, it reassigns the word to a new topic. It repeats this process thousands of times.

5. The Final Output

Eventually, the assignments stabilize. The algorithm then gives you two distinct matrices:

- **Topic-Word Matrix:** A list showing which words define each topic.
- **Document-Topic Matrix:** A list showing the percentage breakdown of topics inside each document.

Conclusion

Topic modeling is a popular natural language processing technique used to create structured data from a collection of unstructured data. In other words, the technique enables businesses to learn the hidden semantic patterns portrayed by a text corpus and automatically identify the topics that exist inside it.

Two popular topic modeling approaches are LSA and LDA. They both seek to discover the hidden patterns in text data, but they make different assumptions to achieve their objective. Where LSA assumes words with similar meanings will appear in similar documents, LDA assumes documents are made up of words that aid in determining the topics.